

Primary Objectives

- Develop a fully interactive 3D photography simulation game using OpenGL
 - Implement realistic camera mechanics including zoom, focus, and lens systems
 - Create smooth player movement and immersive first-person/third-person camera controls
 - Design dynamic environments with animated targets and environmental effects
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Extended Objectives

- Introduce a level-based progression system with increasing difficulty
 - Provide detailed score evaluation based on photography accuracy and positioning
 - Implement unlockable camera lenses and advanced gameplay mechanics
 - Create a gallery system to store and display captured photos and player statistics
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Problem Statement

Many beginner-level OpenGL projects mainly focus on rendering simple 3D objects and basic interactions. These projects often lack engaging gameplay, dynamic environments, advanced camera systems, and interactive user experiences.

This project addresses those limitations by combining computer graphics concepts with game mechanics to create a realistic photography simulation. The game introduces structured progression, intelligent scoring systems, animated targets, and immersive environments that enhance both gameplay and learning experience.

Proposed Solution

The proposed system is a 3D photography adventure game called **Snapshot Safari: The Perfect Shot**. Players explore different environments such as jungles, futuristic cities, and deserts while capturing photographs of moving targets using a virtual camera.

The game evaluates each captured photo based on:

- Camera angle accuracy
- Distance from target
- Lens selection
- Zoom level
- Focus mode

Players progress through multiple levels, unlock advanced lenses, and build a professional photography portfolio throughout the game.

System Features

6.1 Core Features

- Multiple 3D environments including jungle, cyber city, and desert worlds
 - Smooth player movement using keyboard controls
 - First-person and third-person camera system
 - Zoom, focus, and lens-switching functionality
 - Dynamic animated targets such as animals, birds, buildings, and vehicles
 - Real-time HUD (Heads-Up Display) showing score, timer, and gameplay information
 - Weather and lighting variations for immersive gameplay
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6.2 Environment & Level System

The game contains multiple themed levels with unique targets and environmental designs. Each level increases the challenge by introducing different target behaviors, lighting conditions, and gameplay mechanics.

Examples include:

- Jungle wildlife photography
- Futuristic city exploration
- Desert landmark photography

Players must capture enough successful photos to advance to the next level.

6.3 Photo Scoring System

Instead of using a simple fixed score, the game calculates photography quality dynamically. Each photo is evaluated using several components:

- Accuracy of camera alignment
- Distance from the target
- Zoom effectiveness
- Focus mode usage
- Lens compatibility

This system creates a realistic photography experience and demonstrates mathematical calculations in computer graphics and game logic.

6.4 Gallery & Progress Tracking

The game stores all successfully captured photos in an in-game gallery system. Each entry includes:

- Target type
- Environment name
- Lens used
- Zoom level
- Weather condition
- Photo score

The gallery also displays player statistics such as average score and overall performance rating.

System Architecture

Main Components

- OpenGL Rendering Engine
- Game State Management System
- Player & Camera Control System
- Target Animation and Movement System
- Environment Rendering Module
- Dynamic Scoring System
- Gallery & Progress Tracking System
- Weather and Lighting Controller

Execution Flow

1. The game starts with a menu interface
 2. The player selects a photography avatar
 3. The environment and targets are generated
 4. The player explores the 3D world using movement controls
 5. Photos are captured using the camera system
 6. Each photo is evaluated using the scoring algorithm
 7. Progress is checked continuously
 8. New levels and lenses are unlocked based on performance
 9. Final statistics and gallery results are displayed
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Algorithms and Concepts Used

- Trigonometric calculations for camera angles and movement
 - Distance formula for target evaluation
 - Perspective projection using OpenGL
 - Real-time object rendering and transformations
 - Animation using sine and cosine functions
 - State management for game flow control
 - Collision and boundary checking
 - Dynamic score calculation algorithms
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Tools and Technologies

- **Programming Language:** Python
 - **Graphics Library:** OpenGL (GL, GLUT, GLU)
 - **Additional Modules:** math, random, time
 - **Rendering Techniques:** 3D transformations, perspective projection, animation
 - **Development Type:** Real-time interactive graphics application
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Expected Output

- A fully functional 3D photography simulation game
- Smooth player movement and immersive camera controls
- Dynamic environments with animated targets
- Realistic photography scoring system
- Unlockable lenses and gameplay progression
- Interactive gallery showing captured photos and statistics
- Real-time rendering with OpenGL graphics and animations